BSBS • SYSTEM DEVELOPMENT PROJECT

A DIGITAL SOLUTION FOR REPORTING AND TRACKING INFRASTRUCTURE ISSUES IN EDUCATIONAL INSTITUTIONS

A web system for teachers to report infrastructure and equipment faults and for administrators to prioritise, resolve and track them.

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TECH STACK

Java 21 · Spring Boot 3.5 · Thymeleaf · Tailwind CSS · MySQL 8 · · Maven

DURATION

3 months

SUPERVISOR

Resego Tom Rebalone

1 INTRODUCTION & PROBLEM STATEMENT

Background

- Schools depend on infrastructure and equipment that must be reported and repaired.
- Manual tracking causes errors and wasted time (Elaskari et al., 2021).

Problem Statement

- Fault reports are scattered, so status, location and history are unclear.
- Maintenance management falls below best practice (Buys & Nkado, 2006).

2 LITERATURE REVIEW

- Asset management is lifecycle-based, not a static list (ISO 55000).
- Barcodes, QR codes and mobile apps automate asset tracking.
- Gap: faults, location and history in one multi-school system.

Existing work	Strength	Limitation
Elaskari et al. (2021)	Barcodes cut entry errors	Inventory focus only
Chen et al. (2005)	Centralised across depts	Needs WLAN infrastructure
Sulistyo et al. (2022)	QR, history, geolocation	Needs users to scan
Ulum & Sufaidah (2026)	Offline QR inventory	Inventory, not repairs
Proposed artifact	Fault reporting with tracked status, for many schools	

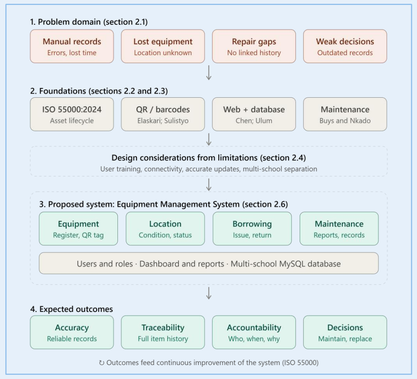


Figure 1. Conceptual framework

3 SYSTEM DESIGN

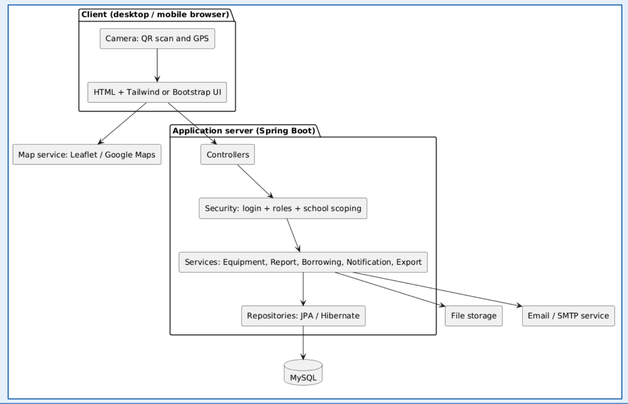
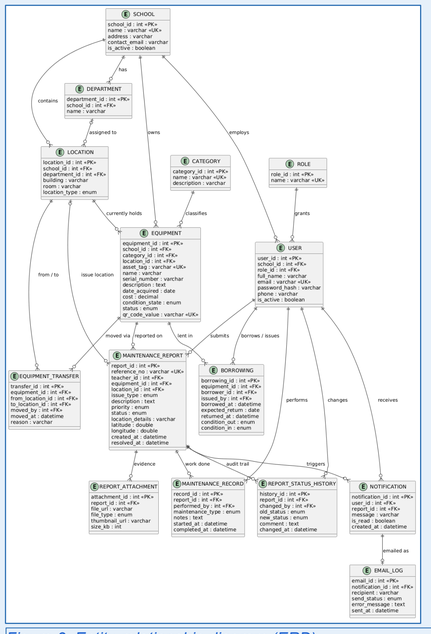


Figure 2. System architecture



- RQ01 Submit reports with category, priority and reference number
- RQ02 Attach photos or video as evidence
- RQ03 Record location, with GPS on mobile
- RQ04 Teacher dashboard to track report status
- RQ05 Admin dashboard to view, filter and manage reports
- RQ06 Update status with a mandatory comment
- RQ07 Email and in-app status notifications
- RQ08 Weekly, monthly and quarterly summary reports
- RQ09 Search previous reports
- RQ10 Export report data as CSV
- RQ11 Highlight urgent reports and count unresolved ones
- Non-functional: HTTPS, 3 s page loads, 50 concurrent users

Figure 3. Entity-relationship diagram (ERD)

4 RESULTS & DISCUSSION

- Show report submission and the admin dashboard in use.
- Evaluate results against requirements RQ01–RQ11.
- Discuss how faults are now reported, tracked and resolved.

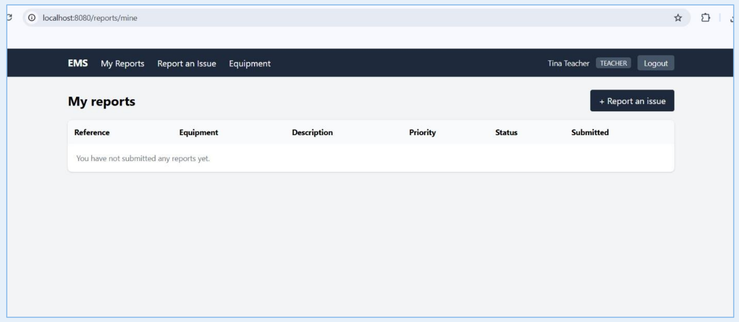


Figure 4. The teacher's "My reports" page in the working system

Evaluation criterion	Method	Target	Result
Functionality	Test cases, RQ01–RQ11	All pass	[ ]
Usability	User testing / SUS survey	[ ]	[ ]
Performance	Page load, 50 users	≤ 3 s	[ ]

5 CONCLUSION

- One central system now records faults, status and history.
- It extends inventory-focused systems with a repair workflow.
- Requirements RQ01–RQ11: [met / partly met — update].

6 RECOMMENDATIONS & FUTURE WORK

- Train teachers and administrators before rollout.
- Limitations: relies on users reporting, and on network access.
- Future: offline mode, RFID tracking and a native mobile app.

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DEMO QR Change Picture

CODE QR Change Picture